

Financial Time Series Lecture 6: Iterative Approaches to Estimating Volatility

Some alternative methods: (Non-parametric methods)

- Moving window estimates
- Use of high-frequency financial data
- Use of daily open, high, low and closing prices (or log prices)

Moving window

A simple approach to capture time-varying feature of the volatility. Hard to determine the size of the window.

Demonstration: Use the `quantmod` package to download the daily trading information of SPDR S&P 500 from January 3, 2003 to April 30, 2017. The tick symbol is `SPY`. Use the adjusted index value to compute daily log returns of SPY. A R script, `mvwindow.R`, is available on the course web.

Instructions:

1. Download the data and save it in your R working directory.
2. Compile the program using the command: `source("mvwindow.R")`
3. To run the program: `mvol=mvwindow(rt,size)`, where “rt” denotes the return series and “size” is the size of the moving window.
4. The output is the volatility, i.e., σ_t , stored in `sigma.t`.

Demonstration shown in class.

Use of High-Frequency Data

Suppose we like to estimate the monthly volatility of a stock return.

Data: Daily returns

Let r_t^m be the t -th month log return.

Let $\{r_{t,i}\}_{i=1}^n$ be the daily log returns within the t -th month.

Using properties of log returns, we have

$$r_t^m = \sum_{i=1}^n r_{t,i}.$$

Assuming that the conditional variance and covariance exist, we have

$$\text{Var}(r_t^m | F_{t-1}) = \sum_{i=1}^n \text{Var}(r_{t,i} | F_{t-1}) + 2 \sum_{i < j} \text{Cov}[(r_{t,i}, r_{t,j}) | F_{t-1}],$$

where F_{t-1} = the information available at month $t - 1$ (inclusive).

Further simplification is possible under additional assumptions.

If $\{r_{t,i}\}$ is a white noise series, then

$$\text{Var}(r_t^m | F_{t-1}) = n \text{Var}(r_{t,1}),$$

where $\text{Var}(r_{t,1})$ can be estimated from the daily returns $\{r_{t,i}\}_{i=1}^n$ by

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n (r_{t,i} - \bar{r}_t)^n}{n - 1},$$

where $\bar{r}_t$ is the sample mean of the daily log returns in month t (i.e., $\bar{r}_t = \sum_{i=1}^n r_{t,i} / n$).

The estimated monthly volatility is then

$$\hat{\sigma}_m^2 = \frac{n}{n - 1} \sum_{i=1}^n (r_{t,i} - \bar{r}_t)^2 \approx \sum_{i=1}^n (r_{t,i} - \bar{r}_t)^2.$$

If $\{r_{t,i}\}$ follows an MA(1) model, then

$$\text{Var}(r_t^m | F_{t-1}) = n \text{Var}(r_{t,1}) + 2(n - 1) \text{Cov}(r_{t,1}, r_{t,2}),$$

which can be estimated by

$$\hat{\sigma}_m^2 = \frac{n}{n - 1} \sum_{i=1}^n (r_{t,i} - \bar{r}_t)^2 + 2 \sum_{i=1}^{n-1} (r_{t,i} - \bar{r}_t)(r_{t,i+1} - \bar{r}_t).$$

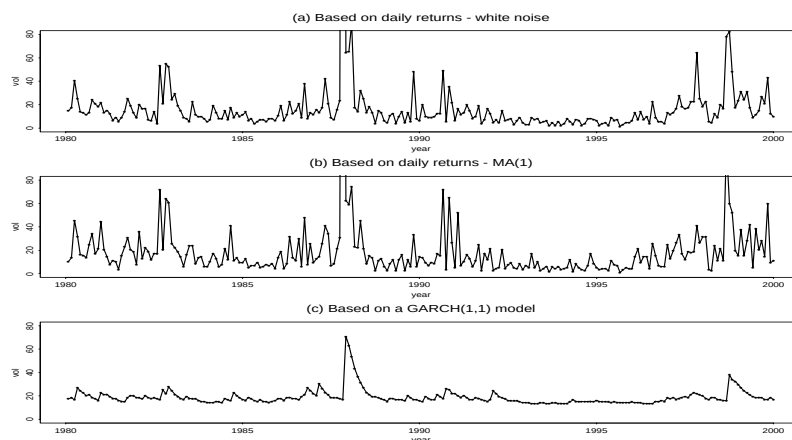


Figure 1: Time plots of estimated monthly volatility for the log returns of S&P 500 index from January 1980 to December 1999: (a) assumes that the daily log returns form a white noise series, (b) assumes that the daily log returns follow an MA(1) model, and (c) uses monthly returns from January 1962 to December 1999 and a GARCH(1,1) model.

Advantage: Simple

Weaknesses:

- Models for daily returns $\{r_{t,i}\}$ are unknown.
- Typically, 21 or 22 trading days in a month, resulting in a small sample size.

See Figure 1 for an illustration; Ex 3.6 of the text.

Realized integrated volatility

If the sample mean $\bar{r}_t$ is zero, then $\hat{\sigma}_m^2 \approx \sum_{i=1}^n r_{t,i}^2$.

⇒ Use cumulative sum of squares of daily log returns within a month as an estimate of monthly volatility.

Consider tick-by-tick data: Apply the idea to *intraday log returns* and obtain realized integrated volatility.

Assume daily log return $r_t = \sum_{i=1}^n r_{t,i}$. The quantity

$$RV_t = \sum_{i=1}^n r_{t,i}^2,$$

is called the *realized* volatility of r_t .

Advantages: simplicity and using intraday information

Weaknesses:

- Effects of market micro-structure noises
- Overlook overnight volatilities.

Further discussion

1. In-filled asymptotic argument. Let Δ be the sampling interval, as $\Delta \rightarrow 0$, the sample size goes to infinity.

Under the assumption that the Δ -interval log returns, e.g. 5-minute returns, are independent and identically distributed, then $\sum_{j=1}^n r_{t,j}^2$ converges to the variance of the daily log return r_t . (Quadratic variation)

2. In practice, however, there are micro-structure noises that affect the estimate such as the bid-ask bounce. In fact, it can be shown that as Δ goes to zero, the observed sum of squares of Δ -interval returns goes to infinity.

What next? Two approaches have been proposed:

- (a) Optimal sampling interval: Bandi and Russell (2006). Find an optimal Δ . Or equivalently, the optimal sample size n^*

$= 6.5 \text{ hours}/\Delta$ can be chosen as

$$n^* \approx \left[\frac{Q}{(\hat{\sigma}_{noise}^2)^2} \right]^{1/3},$$

where $Q = \frac{M}{3} \sum_{j=1}^M r_{t,j}^4$ and $\hat{\sigma}_{noise}^2 = \frac{1}{M} \sum_{j=1}^M r_{t,j}^2$, where M is the number of daily quotes available for the underlying stock and the returns $r_{t,j}$ are computed from the mid-point of the bid and ask quotes.

- (b) Sub-sampling: Zhang et al. (2006). Choose Δ between 10 to 20 minutes. Compute integrated volatility for each of the possible Δ -interval return series. Then, compute the average. In fact, the authors propose a so-called two scales realized volatility (TSRV) estimate. The form is

$$RV = a_n \times ARV_K - b_n \times ARV_J,$$

where ARV_i denotes the average realized volatility of time interval i , a_n is a real number approaching 1 and $b_n = a_n \times n_K/n_J$, and $n_K = (n - K + 1)/K$ with n is the number of transactions within the day. J can be 1 or $J \ll K$. When $J = 1$, the second term can be regarded as estimate of the noise. When K is much larger than J , the second term is typically small.

Use of Daily Open, High, Low and Close Prices

Figure 2 shows a time plot of price versus time for the t th trading day. Define

- C_t = the closing price of the t th trading day;
- O_t = the opening price of the t th trading day;

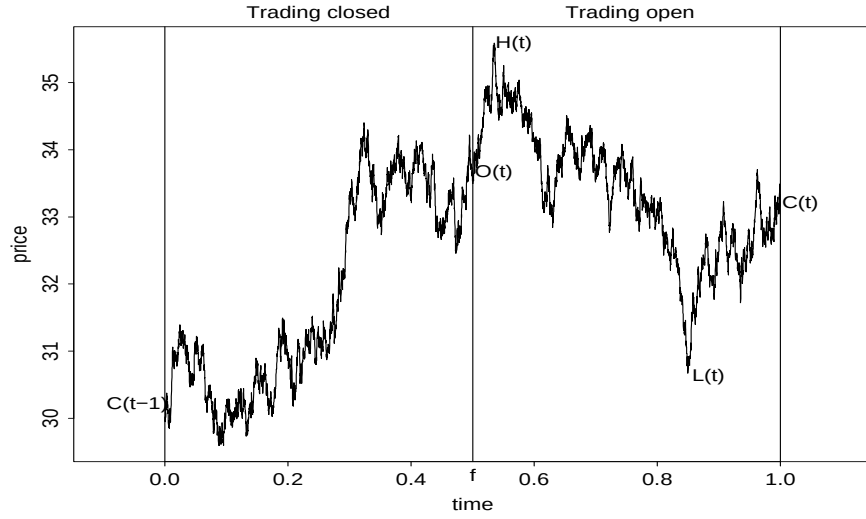


Figure 2: Time plot of price over time: scale for price is arbitrary.

- f = fraction of the day (in interval $[0,1]$) that trading is closed;
- H_t = the highest price of the t th trading period;
- L_t = the lowest price of the t th trading period;
- F_{t-1} = public information available at time $t - 1$.

The conventional variance (or volatility) is $\sigma_t^2 = E[(C_t - C_{t-1})^2 | F_{t-1}]$.
Some alternatives:

- $\hat{\sigma}_{0,t}^2 = (C_t - C_{t-1})^2$;

- $\hat{\sigma}_{1,t}^2 = \frac{(O_t - C_{t-1})^2}{2f} + \frac{(C_t - O_t)^2}{2(1-f)}, \quad 0 < f < 1;$
- $\hat{\sigma}_{2,t}^2 = \frac{(H_t - L_t)^2}{4 \ln(2)} \approx 0.3607(H_t - L_t)^2;$
- $\hat{\sigma}_{3,t}^2 = 0.17 \frac{(O_t - C_{t-1})^2}{f} + 0.83 \frac{(H_t - L_t)^2}{(1-f)4 \ln(2)}, \quad 0 < f < 1;$
- $\hat{\sigma}_{5,t}^2 = 0.5(H_t - L_t)^2 - [2 \ln(2) - 1](C_t - O_t)^2,$
which is $\approx 0.5(H_t - L_t)^2 - 0.386(C_t - O_t)^2;$
- $\hat{\sigma}_{6,t}^2 = 0.12 \frac{(O_t - C_{t-1})^2}{f} + 0.88 \frac{\hat{\sigma}_{5,t}^2}{1-f}, \quad 0 < f < 1.$

A more precise, but complicated, estimator $\hat{\sigma}_{4,t}^2$ was also considered. But it is close to $\hat{\sigma}_{5,t}^2$.

Defining the efficiency factor of a volatility estimator as

$$\text{Eff}(\hat{\sigma}_{i,t}^2) = \frac{\text{Var}(\hat{\sigma}_{0,t}^2)}{\text{Var}(\hat{\sigma}_{i,t}^2)},$$

Garman and Klass (1980) found that $\text{Eff}(\hat{\sigma}_{i,t}^2)$ is approximately 2, 5.2, 6.2, 7.4 and 8.4 for $i = 1, 2, 3, 5$ and 6, respectively, for the simple diffusion model entertained.

For log-return volatility, one takes the logarithms of the Open, High, Low and Close prices.

Define

- $o_t = \ln(O_t) - \ln(C_{t-1})$ be the normalized open;
- $u_t = \ln(H_t) - \ln(O_t)$ be the normalized high;
- $d_t = \ln(L_t) - \ln(O_t)$ be the normalized low;
- $c_t = \ln(C_t) - \ln(O_t)$ be the normalized close.

Suppose that there are n days of data available and the volatility is constant over the period. Yang and Zhang (2000) recommend the estimate

$$\hat{\sigma}_{yz}^2 = \hat{\sigma}_o^2 + k\hat{\sigma}_c^2 + (1 - k)\hat{\sigma}_{rs}^2$$

as a robust estimator of the volatility, where

$$\begin{aligned}\hat{\sigma}_o^2 &= \frac{1}{n-1} \sum_{t=1}^n (o_t - \bar{o})^2 \quad \text{with} \quad \bar{o} = \frac{1}{n} \sum_{t=1}^n o_t, \\ \hat{\sigma}_c^2 &= \frac{1}{n-1} \sum_{t=1}^n (c_t - \bar{c})^2 \quad \text{with} \quad \bar{c} = \frac{1}{n} \sum_{t=1}^n c_t, \\ \hat{\sigma}_{rs}^2 &= \frac{1}{n} \sum_{t=1}^n [u_t(u_t - c_t) + d_t(d_t - c_t)], \\ k &= \frac{0.34}{1.34 + (n+1)/(n-1)}.\end{aligned}$$

This estimate seems to perform reasonably well.

Remark: One must consider the stock split in the above calculation.

Some work using daily range. For log returns, daily range is defined as

$$r_t = \ln(H_t) - \ln(L_t).$$

This is related to the **duration models** to be discussed later in high-frequency data.

Takeaway

Some alternative approaches to volatility estimation are currently under intensive study. It is rather early to assess the impact of these methods. It is a good idea in general to use more information. However, regulations and institutional effects need to be considered.

Long Memory Processes (LMPs)

- ACF of ARMA, which is an ARIMA($p, d = 0, q$), converges to zero at an exponential rate, i.e., for large h :

$$\rho_h \approx \frac{1}{c^h}, \quad |c| > 1.$$

(E.g., recall that for an AR(1) $\rho_h = \phi^h$, where $|\phi| < 1$.)

- LMPs have a slower ACF decay (hyperbolic rate), i.e., for large h :

$$\rho_h \approx \frac{1}{h^\alpha}, \quad 0 < \alpha < 1.$$

- One can equivalently define LMPs in terms of the *Hurst coefficient*:

$$H = 1 - \frac{\alpha}{2}, \quad \implies \quad \frac{1}{2} \leq H \leq 1.$$

- Incorporate LM into conditional mean μ_t of the zero-mean process $\{r_t\}$ via a FARIMA(p, d_m, q), a *fractionally integrated ARMA*:

$$\phi_m(B)(1 - B)^{d_m}r_t = \theta(B)a_t,$$

where $\{a_t\} \sim \text{WN}(0, \sigma_a^2)$, $d_m \in \mathbb{R}$, and $\phi_m(z)$ and $\theta(z)$ are AR(p) and MA(q) polynomials. A FARIMA (also called ARFIMA) is stationary if and only if $|d_m| < 0.5$, and there are 3 cases to consider:

If $-0.5 < d_m < 0$: $r_t \sim \text{AR}(\infty)$, has short memory (anti-persistent or mean-reverting).

If $d_m = 0$: $r_t \sim \text{ARMA}$, also has short memory.

If $0 < d_m < 0.5$: r_t has long memory (is an LMP), in which case d_m can be related to H as:

$$d_m = H - \frac{1}{2}.$$

- Can estimate and test for LM. The R/S statistic was originally proposed for this, but there are several other (ongoing) methods.

- Incorporate LM into conditional variance σ_t^2 of the zero-mean process $\{r_t\}$ via a FIGARCH(p, d_v, q), a *fractionally integrated GARCH*:

$$\phi_v(B)(1 - B)^{d_v} a_t = \omega + (1 - \beta(B))\sigma_t^2,$$

where $\{a_t\} \sim \text{WN}(0, \sigma_a^2)$, $0 < d_v < 1$, $\phi_v(z)$ is an AR(p) polynomial, and

$$\beta(z) = \sum_{j=1}^q \beta_j z^j.$$

(Note: this is a special case of an IGARCH(p, q); incorporating LM into other types of GARCH is a subject of ongoing research...)

Example 1 (USO Returns). *Analyze USO stock data (energy market). As is typical, there is little serial correlation...(Fig 1).*

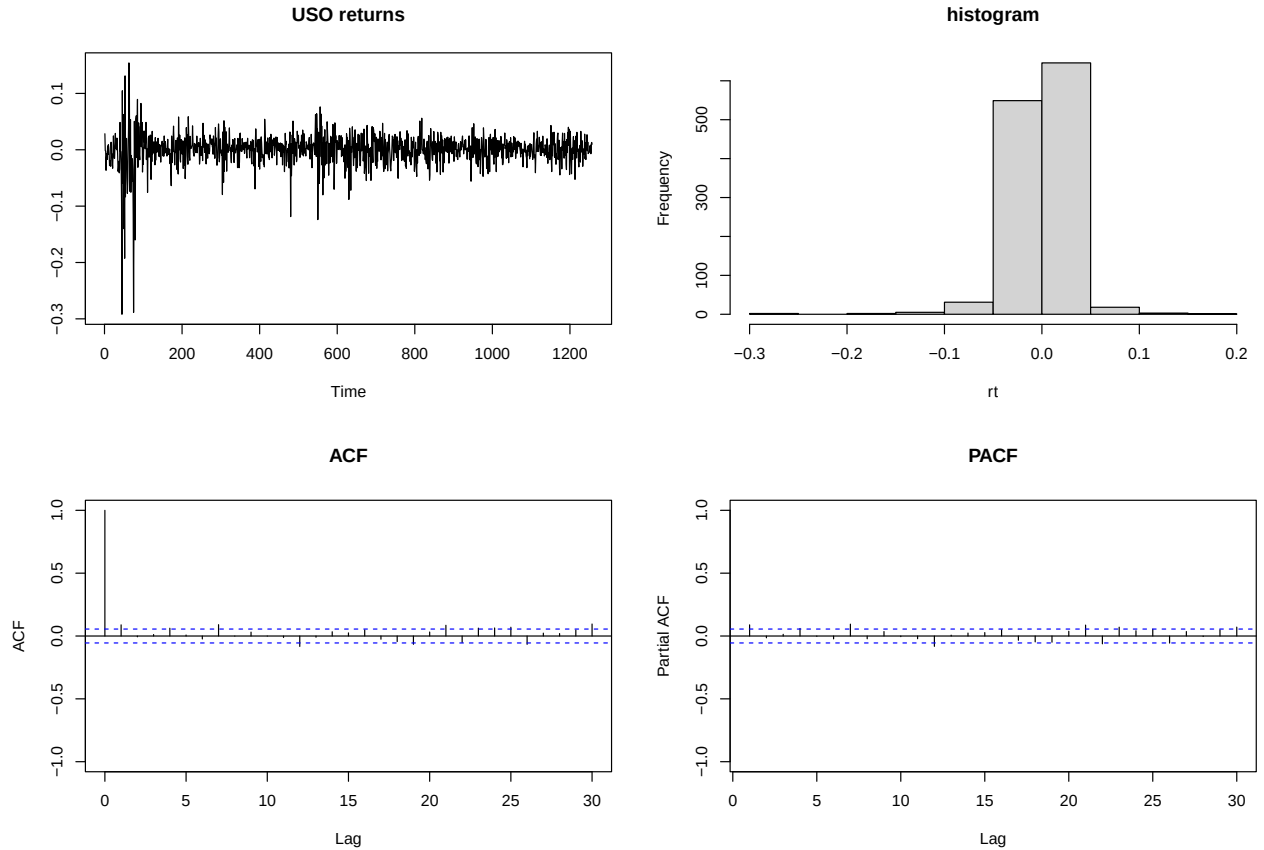


Figure 1: USO stock returns from 2020-01-01 to 2025-01-01 (daily closing prices).

However, estimation of the Hurst coefficient from package *pracma* yields values of 0.63 (R/S statistic) and 0.66 (empirical), suggesting some LM is present.

Searching for a $FARIMA(p, d_m, q)$ - $FIGARCH(1, d_v, 1)$ suggests optimal values of $(p, q) = (2, 0)$ and $d_m = 0 = d_v$, so that an overall fitted model with no LM cannot be rejected!

```
#####
### Code for Example 1: Get USO data, compute log returns, basic plotting.
> library(quantmod)
> getSymbols("USO", src = "yahoo", from = "2020-01-01", to = "2025-01-01")
> USO.rt = diff(log(USO$USO.Adjusted))
> rt = as.vector(USO.rt)[-1]
> par(mfrow=c(2,2))
> ts.plot(rt, main="USO returns", ylab="")
> hist(rt, main="histogram")
> acf(rt, main="ACF")
> pacf(rt, main="PACF")

### Perform R/S analysis to estimate the Hurst exponent
> library(pracma)
> rs_results <- hurstexp(rt, display = TRUE)
> print(rs_results$Hrs) # Corrected R/S Hurst exponent
> print(rs_results$He)  # Empirical Hurst exponent

Corrected R over S Hurst exponent:    0.6272975
Empirical Hurst exponent:             0.6550958

### Search for an appropriate FARIMA
### Note: the autoarfima function is richer than the auto.arima
###       ("forecast" package), since it will also look for the
###       optimal value of fractional differencing (FARIMA).
> library(rugarch)
> auto.fit <- autoarfima(
  data = rt,                # Your time series data
  ar.max = 5,               # Maximum autoregressive (p) order to check
  ma.max = 5,               # Maximum moving average (q) order to check
  criterion = "BIC",        # Optimization metric: "AIC", "BIC", "SIC", or "HQIC"
  arfima = NULL, include.mean = NULL,
  distribution.model = "sstd",
  method = "full"           # "full" estimates all combos; "partial" is faster
)
> print(auto.fit)
```

```

*-----*
*           ARFIMA Model Fit           *
*-----*

```

```

Mean Model : ARFIMA(2,0,0)
Distribution : sstd

```

Optimal Parameters

```

-----
      Estimate  Std. Error  t value Pr(>|t|)
ar1    0.000000         NA         NA      NA
ar2   -0.119066    0.025187   -4.7273   2e-06
sigma  0.027301    0.001574   17.3478   0e+00
skew   0.827135    0.029805   27.7516   0e+00
shape  3.336840    0.318752   10.4685   0e+00

```

Robust Standard Errors:

```

      Estimate  Std. Error  t value Pr(>|t|)
ar1    0.000000         NA         NA      NA
ar2   -0.119066    0.033645   -3.5389 0.000402
sigma  0.027301    0.001725   15.8304 0.000000
skew   0.827135    0.034182   24.1981 0.000000
shape  3.336840    0.344174    9.6952 0.000000

```

LogLikelihood : 2944.093

Information Criteria

```

-----
Akaike          -4.6780
Bayes           -4.6616
Shibata         -4.6780
Hannan-Quinn   -4.6718

```

Fit the suggested ARFIMA(2,d,0)-FIGARCH(1,d,1)

```

> my.spec <- ugarchspec(
  mean.model      = list(armaOrder = c(2, 0), arfima = TRUE, include.mean = F),
  variance.model  = list(model = "fiGARCH", garchOrder = c(1, 1)),
  distribution.model = "sstd"
)
> fit <- ugarchfit(spec = my.spec, data = rt)
> print(fit)

```

Conditional Variance Dynamics

GARCH Model : fiGARCH(1,1)

Mean Model : ARFIMA(2,d,0)

Distribution : sstd

Optimal Parameters

	Estimate	Std. Error	t value	Pr(> t)
ar1	0.015170	0.048969	0.309787	0.756723
ar2	-0.053289	0.033284	-1.601022	0.109372
arfima	0.000000	0.043928	0.000001	0.999999
omega	0.000013	0.000003	3.824796	0.000131
alpha1	0.000000	0.101851	0.000002	0.999998
beta1	0.859058	0.040135	21.403942	0.000000
delta	1.000000	0.132882	7.525487	0.000000
skew	0.775270	0.036594	21.185748	0.000000
shape	6.518451	1.374101	4.743792	0.000002